

Bit-wise Binary Addition Using Full Adders and Two's Complement Subtraction

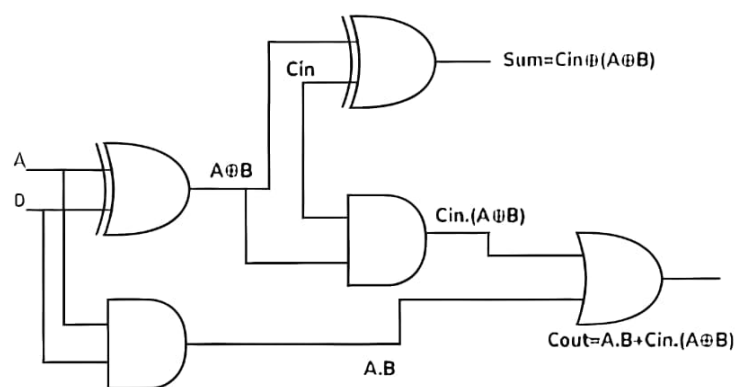
Name: Abhijit Santra

Roll No.: CRS2502

Introduction

Binary arithmetic is the fundamental operation performed by digital computers. A **Full Adder** is a combinational logic circuit that adds two binary bits along with an input carry, making it the basic building block of Arithmetic Logic Units (ALUs).

Full Adder



Inputs			Outputs	
A	B	C	Sum	Carry
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Figure 1: Full Adder Circuit and Truth Table

Bit-wise Addition Using Full Adders

A full adder accepts three inputs: A , B , and C_{in} , producing:

$$\text{Sum} = A \oplus B \oplus C_{in}$$

$$C_{out} = AB + C_{in}(A \oplus B)$$

For multi-bit addition, several full adders are cascaded to form a **Ripple Carry Adder**. The carry generated by each stage is forwarded to the next higher bit until the most significant bit is processed.

- LSB receives $C_{in} = 0$.
- Each stage computes one sum bit and propagates the carry.
- The final carry indicates arithmetic overflow for unsigned numbers.

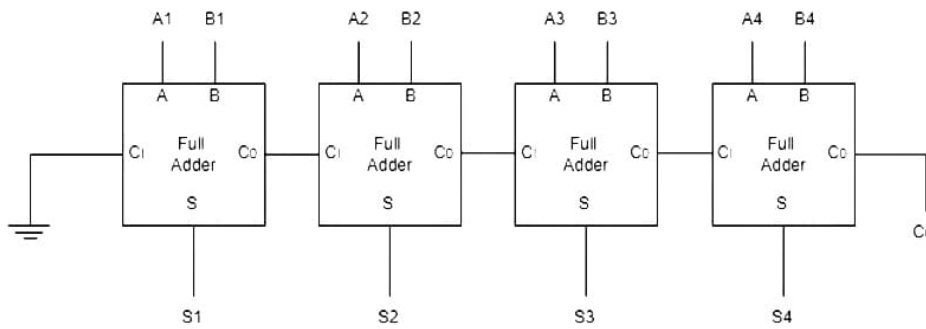


Figure 2: 4-bit Ripple Carry Adder

Subtraction Using Two's Complement

Binary subtraction utilizes the same adder hardware by converting the subtrahend into its two's complement (invert all bits and add 1). The resulting value is then added to the minuend using the ripple carry adder. This eliminates the need for dedicated subtraction circuitry and improves hardware efficiency.

Conclusion

Full adders enable efficient bit-wise arithmetic by cascading simple logic blocks into ripple carry adders. The same architecture performs both addition and subtraction through two's complement representation, providing a compact, scalable, and cost-effective implementation widely used in modern digital processors.

References

1. OpenAI, *ChatGPT (GPT-5.5)*, OpenAI, 2026. Available: <https://chatgpt.com>
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